Aleksandr Movsisian

ETH Zurich, Zurich, Switzerland

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Education

Lomonosov Moscow State University (MSU) **Sep 2023 – Jun 2025**
Master of Science in Physics, major coursework GPA: 5.0/5.0 *Moscow, Russia*
Specialization: Quantum and Nonlinear Oscillatory Systems

- **Thesis title:** Standard quantum limit overcoming in the broadband variational measurement

Lomonosov Moscow State University (MSU) **Sep 2019 – Jun 2023**
Bachelor of Science in Physics, major coursework GPA: 4.53/5.0 *Moscow, Russia*

- **Thesis title:** Ultimate sensitivity of quantum optical sensors

Research Experience

ETH Zurich, Photonics Lab **Jul 2026 – Present**
Research Assistant, Supervisor: Prof. Lukas Novotny *Zurich, Switzerland*

KAUST, Integrated Photonics Lab **Sep 2025 – Jul 2026**
Research Assistant, Supervisor: Prof. Yating Wan *Thuwal, Saudi Arabia*

- Implemented weight-matrix compression algorithm for ViT/CNN architectures, benchmarked the accuracy-compression tradeoff across six datasets (MNIST, FMNIST, CIFAR-10, ImageNet, RESISC45, CheXpert), and demonstrated the compressed CIFAR-10 model on a photonic device on LN
- Suggested and executed calibration workflow for a reconfigurable 10×10 MZI mesh on SOI, including 50-channel voltage source calibration and automated measurement pipelines
- Optimized racetrack filters on LNOI for parallel optical computing and characterize their perfomance
- Ran eigenmode and FDTD simulations to optimize ring-to-bus coupling across critically coupled, overcoupled, and undercoupled regimes
- Investigated a system of two coupled active resonators to observe exceptional-point (EP) behavior, and characterized QD lasers in the EP regime

MSU, Quantum and Precision Measurements Group **Feb 2022 – Jul 2025**
Research Assistant, Supervisor: Prof. Sergey P. Vyatchanin *Moscow, Russia*

- Conducted a theoretical analysis of broadband multidimensional variational readout in a triply resonant cavity optomechanical system, accounting for asymmetric coupling and loss, and quantifying the resulting quantum noise and force sensitivity
- Analyzed the impact of optical loss and intracavity squeezing on the quantum-noise budget for force measurement in the triply resonant optomechanical scheme, comparing single- and two-photon squeezing
- Studied the Standard Quantum Limit in resonant optomechanical force sensing and evaluated SQL-beating strategies based on variational measurements and injected squeezed states of light

EPFL, Laboratory of Photonic Integrated Circuits and Quantum Measurements **Jun 2024 – Sep 2024**
Research Intern, Supervisor: Prof. Tobias J. Kippenberg *Lausanne, Switzerland*

- Optimized design of a thin-film PPLN circuit for the light squeezing and estimated the noise reduction level
- Designed and tested low-pass filters for LNOI vacuum squeezers
- Assembled a free-space balanced homodyne setup, performed periodic poling on lithium niobate samples
- Determined optimal parameters for an on-chip corrugated Fabry-Perot cavity on LNOI

Russian Quantum Center, Group of Coherent Microoptics and Radiophotonics **Sep 2023 – Oct 2023**
Supervisor: Prof. Igor A. Bilenko *Moscow, Russia*

- Theoretically investigated the generation of squeezed states in microring resonators and the use of dual-pumped systems to enhance squeezing level in SiN platforms

National Research Center "Kurchatov Institute" **Sep 2021 – Dec 2021**
Project intern *Moscow, Russia*

- Assembled an experimental setup for solving the inverse problem of diffraction on unstructured objects and conducted measurements

Core skills & Coursework

- **Programming skills:** Python, MATLAB, HPC (Slurm)
- **Specialized software:** Lumerical, COMSOL, Origin, Wolfram Mathematica, Klayout
- **Other programs:** LaTeX, Adobe Illustrator, Figma, MS Office
- **English proficiency:** IELTS 7.0, TOEFL 95 (2024)
- **Advanced courses:** Quantum theory of measurements, Oscillatory systems with small dissipation, Distributed oscillating systems, Low dissipation oscillating systems, Parametric and self-oscillating systems, Quantum communications and computations, Devices and methods of precision measurements based on quantum effects, Fluctuation processes and signal processing in measuring systems,

Publications

- **A. A. Movsisian**, A. I. Nazmiev, A. B. Matsko, S. P. Vyatchanin, "Noiseless signal amplification in an opto-mechanical transducer," Opt. Lett. 50, 5262-5265 (**2025**), LIGO-P2400323
- **A. A. Movsisian**, S. P. Vyatchanin, "Squeezing for Broadband Multidimensional Variational Measurement," Journal of the Optical Society of America B. Vol. 41, Issue 9, pp. 2003-2013 (**2024**), LIGO-P2300325

Conferences

- **A.A. Movsisian**, S.P. Vyatchanin, "Using a quadratically nonlinear environment to increase sensitivity in broadband multidimensional variational measurement", The 66th MIPT Scientific Conference, Moscow, Russia, 2024
- **A.A. Movsisian**, S.P. Vyatchanin, "Squeezing for Broadband Multidimensional Variational Measurement", International Scientific Conference for Young Scientists "Lomonosov", Moscow, Russia, 2024

Awards & Scholarships

- The Graduate Dean's Scholar Award at UCLA — 2025
  Merit-based recruitment fellowship for outstanding incoming students
- EPFL Excellence in Engineering, Student Summer Fellowship — 2024
  Awarded to 40 out of 2060 applicants.
- Winner of the V. Potanin personal scholarship competition — 2024
  One of 750 students selected nationwide in Russia from 6,416 applicants.
- Grant support through the "Leader" competition for theoretical research in fundamental physics from the Foundation for Assistance to the Development of Theoretical Physics and Mathematics, "BASIS" — 2023
  Awarded among all Russian theoretical scientific groups.
- Winner of the "Lomonosov" Universiade in Theoretical and Applied Physics — 2023
- Increased state academic scholarship for outstanding achievements in social activities — 2020

Teaching, Academic Service & Extracurricular Activities

- Ad hoc reviewer, Applied Physics Reviews (AIP Publishing) — Jan 2026
- Ad hoc reviewer, Journal of the Optical Society of America B (JOSA B) — Feb 2026
- Lecturer at the MSU Project School for high school students — Aug 2021 - Aug 2021
- Mentor for Physics at Foxford Online School — Aug 2020 - Jan 2022
- Elected member of the Student Council of the Faculty of Physics of MSU — Oct 2019 - Oct 2023
- Administrative Assistant at MSU's Youth Research Platform — Nov 2020 - Nov 2021
  Assisted in launching a competition aimed at supporting emerging applied scientific ideas, securing approximately $90,000 in funding, and organizing communication between contestants and jurors.
- Assistant at MSU's Youth Research Workshop — Feb 2023 - Feb 2023
  Led the online promotion of a scientific school aimed at supporting young scientists.